

The prevention and control the type-2 diabetes by changing lifestyle and dietary pattern

Mohammad Asif

Department of Pharmacy, GRD Institute of Management and Technology, Dehradun, (Uttarakhand), India

ABSTRACT

Type-2 diabetes is a major, non-communicable disease with increasing prevalence at a global level. Type-2 diabetes results when the body does not make enough insulin or the body cannot use the insulin it produces. Type-2 diabetes is the leading cause of premature deaths. Improperly managed, it can lead to a number of health issues, including heart diseases, stroke, kidney disease, blindness, nerve damage, leg and foot amputations, and death. Type-2 diabetes or adult-onset diabetes is most common type of diabetes, usually begins when a person is in his or her mid-50s, but diabetes is not inevitable. Minor changes in your lifestyle can greatly reduce your chances of getting this disease. Therefore, in order to prevent this condition, action should be taken regarding the modifiable factors that influence its development-lifestyle and dietary habits. However, with proper testing, treatment and lifestyle changes, healthy eating as a strategy, promote walking, exercise, and other physical activities have beneficial effects on human health and prevention or treatment of diabetes, promoting adherence to this pattern is of considerable public health importance.

Key words: Diet, lifestyle, non-communicable disease, public health, type-2 diabetes

INTRODUCTION

Diabetes mellitus or type-2 diabetes, is one of the major non-communicable and fastest growing public health problems in the world, is a condition difficult to treat and expensive to manage. It has been estimated that the number of diabetes sufferers in the world will double from the current value of about 190 million to 325 million during the next 25 years.^[1-3] Individuals with type-2 diabetes are at a high risk of developing a range of debilitating complications

such as cardiovascular disease, peripheral vascular disease, nephropathy, changes to the retina and blindness that can lead to disability and premature death. It also imposes important medical and economic burdens. Genetic susceptibility and environmental influences seem to be the most important factors responsible for the development of this condition. However, a drastic increase of physical inactivity, obesity, and type-2 diabetes has been recently observed. The fact indicates that obesity and physical inactivity may constitute the main reasons for the increasing burden of diabetes in the developed world.^[4-10]

Fortunately, because environmental factors are modifiable, disease manifestation from these factors is largely preventable. Diet is one of the major factors now linked to a wide range of diseases including diabetes. The amount and type of food consumed is a fundamental determinant of human health. Diet constitutes a crucial aspect of the overall management of diabetes, which may involve diet alone, diet with oral hypoglycemic drugs, or diet with insulin.^[11-15] Diet is individualized depending on age, weight, gender, health condition, and occupation etc. The dietary guidelines as used in this review are sets of advisory statements that give quick dietary advice for the management of the diabetic population

Address for correspondence: Mr. Mohammad Asif,
Department of Pharmacy, GRD Institute of Management and Technology,
Dehradun, (Uttarakhand) - 248 009, India.
E-mail: aasif321@gmail.com

Access this article online

Quick Response Code:



Website:
www.jehp.net

DOI:
10.4103/2277-9531.127541

Copyright: © 2014 Asif M. This is an open-access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

This article may be cited as: Asif M. The prevention and control the type-2 diabetes by changing lifestyle and dietary pattern. J Edu Health Promot 2014;3:1.

in order to promote overall nutritional well-being, glycogenic control, and prevent or ameliorate diabetes-related complications.

Objectives of dietary treatment of diabetes

The aims of dietary treatment of diabetes are:

- To achieve optimal blood glucose concentrations.
- To achieve optimal blood lipid concentrations.
- To provide appropriate energy for reasonable weight, normal growth, and development, including during pregnancy and lactation.
- To prevent, delay, and treat diabetes-related complications.
- To improve health through balanced nutrition.

The attempts to adhere to the conventional food measurements in order to comply with prescriptions of the so-called 'diabetic diet' usually result in unnecessary restrictions, overindulgence, or monotonous consumption of certain food items, e.g., unripe plantain/beans. This is a consequence of illiteracy, poverty, and cultural misconceptions about the role of diet in the management of diabetes. This is usually the most problematic aspect of diabetes care. The usually recommended daily energy intake for the non-obese diabetic patient is between 1500 and 2500 calories per day, the average allowance being 2000 k calories per day. The recommendation for the overweight diabetic patient is between 800 and 1500 k calories per day, while the underweight (including growing children and adolescents) should be allowed at least 2500 k calories/day.^[16,17]

DIET AND DIABETES

The beneficial effect of the dietary pattern on diabetes mellitus and glucose metabolism in general and traditional food pattern was associated with a significant reduction in the risk of developing type-2 diabetes. The dietary pattern emphasizes a consumption of fat primarily from foods high in unsaturated fatty acids, and encourages daily consumption of fruits, vegetables, low fat dairy products and whole grains, low consumption of fish, poultry, tree nuts, legumes, very less consumption of red meat.^[18-20] The composition of diet is one of the best known dietary patterns for its beneficial effects on human health that may act beneficially against the development of type-2 diabetes, including reduced oxidative stress and insulin resistance. High consumption of vegetables, fruits, legumes, nuts, fish, cereals and oil leads to a high ratio of monounsaturated fatty acids to saturated fatty acids, a low intake of trans fatty acids, and high ingestion of dietary fiber, antioxidants, polyphenols. The diets are characterized by a low degree of energy density overall; such diet prevent weight gain and exert a protective effect on the development of type-2 diabetes, a condition that is partially mediated through weight maintenance. Greater adherence to the diet in combination with light physical activity was associated with lower odds of having diabetes after adjustment for various factors.^[21-25] On the other hand, a paleolithic diet (i.e., a diet consisting of lean meat, fish, shellfish, fruits and vegetables,

roots, eggs and nuts, but not grains, dairy products, salt or refined fats, and sugar) was associated with marked improvement of glucose tolerance while control subjects who were advised to follow a diet did not significantly improve their glucose tolerance despite decreases in weight and waist circumference.^[26-28] People most likely to get diabetes are: People who are overweight, upper-body obesity, have a family history of diabetes, age 40 or older, and women (50% more often than men).

TREATMENTS OF DIABETES

Each person needs individualized treatment. Type-1 diabetes always requires insulin, diet, and exercise. Type-2 diabetics require insulin or oral hypoglycemic agents (medication that helps lower blood sugar), if diet and exercise alone fail to lower blood glucose. If you have diabetes, you need to have a medical team (doctor, nutritionist, and health educator or nurse) working with you. Whichever type of diabetes you have, the key to proper control is balancing the glucose and the insulin in the blood. This means adjusting your diet, activity, and sometimes taking medication.^[29,30]

GENERAL DIETARY GUIDELINES

Modern dietary management of diabetes essentially involves modifications of the quality and quantity of food to be taken by the diabetic patient. The following guidelines are applicable to diabetes irrespective of type, weight status, age, gender, or occupation.^[31-40]

- Most of the carbohydrate consumed should be in the form of starch (polysaccharides) such as maize, rice, beans, bread, potatoes etc.
- All refined sugars such as glucose, sucrose, and their products (soft drinks, sweets, toffees, etc.) and honey should be avoided, except during severe illness or episodes of hypoglycemia. These foods contain simple sugar, which is easily absorbed causing rapid rise in blood sugar.
- Non-nutritive sweeteners, e.g., Canderel, saccharine, NutraSweet, aspartame are suitable sugar substitutes for diabetic subjects.
- Animal fat such as butter, lard, egg yolk, and other foods high in saturated fatty acids and cholesterol should be reduced to a minimum and be replaced with vegetable oils, particularly polyunsaturated fats.
- Salt should be reduced whether hypertensive or not.
- Protein (fish, meat, beans, crab, crayfish, soyabean, chicken, etc.) and salt are restricted for those with diabetic nephropathy.
- Cigarette smoking should be avoided by diabetic patients. Alcohol should be taken only in moderation.
- The items allowed for free consumption include: Water, green leafy vegetables, tomatoes, onions, cucumber, aubergine, peppers, vegetable salad without cream. Any brand of tea, coffee, or drinks that contain very low or no calories.
- For patients too ill to eat solid food, a fluid or semi-solid

diet should be substituted (papaya, soya bean, custard, etc.).

- Patients treated with insulin or certain oral hypoglycemic agents, e.g., sulfonylureas must be advised to eat regularly and often to prevent hypoglycemia- 3 meals a day plus suitable snacks in between, e.g., fresh fruits.
- Small meals spaced over the day, rather than 1 or 2 big meals, are helpful in avoiding post-prandial peaks in blood sugar.

When overweight diabetic patients drop some weight by trimming down 'serving sizes' and calories, insulin sensitivity improves, thereby optimizing drug therapy. The fundamental principle behind maintenance of body weight is the energy balance. This group should be encouraged to maintain their current weight by: Maintaining current 'serving sizes,' eating about the same amount of food each day, eating at about the same times each day, taking their drugs at the same times each day, and exercising at the same times each day. These patients should also endeavor to choose their daily foods from starches, vegetables, fruits, and protein, while limiting the amount of fats.^[41-45]

Dietary approaches to diabetes

Food can be powerful in preventing and reversing diabetes. However, dietary approaches have changed as we have learned more about the disease. The traditional approach to diabetes focuses on limiting refined sugars and foods that release sugars during digestion-starches, breads, fruits, etc. With carbohydrates reduced, the diet may contain an unhealthy amount of fat and protein. Therefore, diabetes experts have taken care to limit fats- especially saturated fats that can raise cholesterol levels, and to limit protein for people with impaired kidney function. The new approach focuses more attention on fat. Fat is a problem for people with diabetes. The more fat there is in the diet, the harder time insulin has in getting glucose into the cells. Conversely, minimizing fat intake and reducing body fat help insulin do its job much better. Newer treatment programs drastically reduce meats, high-fat dairy products, and oils.^[46-50] At the same time, they increase grains, legumes, fruits, and vegetables. The study found that patients on oral medications and patients on insulin were able to get off of their medications after some days on a near-vegetarian diet and exercise program. During 2 and 3-year follow-ups, most people with diabetes treated with this regimen have retained their gains. The dietary changes are simple, but profound, and they work.^[51-53]

EFFECTS OF THE FRUIT AND VEGETABLES ON THE HUMAN HEALTH

Fruits constitute a commercially important and nutritionally indispensable food commodity. Being a part of a balanced diet, fruits play a vital role in human nutrition by supplying the necessary growth regulating factors essential for maintaining normal health. They have been especially valuable for their ability to prevent vitamin C and vitamin A deficiencies. Fruits and vegetables are good source

of vitamins, minerals, flavonoids (anti-oxidants), saponins, polyphenols, carotenoids (vitamin A-like compounds), isothiocyanates (sulfur-containing compounds), and several types of dietary fibers. The fruits and vegetables not only prevent malnutrition but also help in maintaining optimum health through a host of chemical components that are still being identified, tested, and measured. They prevent various chronic diseases like stroke, hypertension, birth defects, cataracts, diabetes, heart disease, cancers, diverticulosis, obstructive pulmonary disease (asthma and bronchitis), and obesity etc.^[53-56] Diets that are high in insoluble fiber may offer the best protection against this disease. Fruits and vegetables are high in cellulose-a type of insoluble fiber. Diets that are high in fiber may be able to help in the management of diabetes. Soluble fiber delays glucose absorption from the small intestine and thus may help prevent the spike in blood glucose levels that follow a meal or snack. The long-term effect may be insignificant, however, due to the many other factors that affect blood glucose. The effects of the fruit and vegetables on the human health allowed to once again measuring the enormous stakes.^[57-60] More and more emphasis is put on the importance of the diversity of food, and in particular of the fruit and vegetables. This new and effective approach to diabetes is remarkably simple. Here are 4 simple steps to managing your blood sugar (and weight, blood pressure, and cholesterol) with diet.^[61-70]

Begin a vegan diet: Avoid animal products

Animal products contain fat, especially saturated fat, which is linked to heart disease, insulin resistance, and certain forms of cancer. These products also contain cholesterol and, of course, animal protein. It may surprise you to learn that diets high in animal protein can aggravate kidney problems and calcium losses. Animal products never provide fiber or healthful carbohydrates. A *vegan* diet is one that contains no animal products at all. Therefore, you'll have to avoid red meat, poultry, fish, dairy products, and eggs.

Go high fiber

Aim for 40 grams of fiber a day, but start slowly. Load up on beans, vegetables, and fruits. Choose whole grains (barley, oats, millet, whole-wheat, etc.). Aim for at least 3 grams per serving on food labels and at least 10 grams per meal.

Low-fat, vegetarian diets are ideal for people with diabetes

The health benefits of a low-fat vegetarian diet such as portions of vegetables, grains, fruits, and legumes (excluding animal products) in people with type-2 diabetes. The vegan diet is based on American Diabetes Association (ADA) guidelines; the results of this study were astounding: Forty-three percent of the vegan group reduced their diabetes medications. Among those participants who didn't change their lipid-lowering medications, the vegan group also had more substantial decreases in their total and LDL cholesterol levels.

Avoid added vegetable oils and other high-fat foods (avoid)

Although most vegetable oils are in some ways healthier than animal fats, you will still want to keep them to a minimum. All fats and oils are highly concentrated in calories. A gram of any fat or oil contains 9 calories, compared with only 4 calories for a gram of carbohydrate. Avoid foods fried in oil, oily toppings, and olives, avocados, and peanut butter. Aim for no more than 2-3 grams of fat per serving of food, e.g., white or wheat bread, most cold cereals, watermelon, pineapple, baking potatoes, sugar.

Favor foods with a low glycemic index (enjoy)

The glycemic index identifies foods that increase blood sugar rapidly. This handy tool allows you to favor foods that have much less effect on blood sugar. High-glycemic-index foods include sugar itself, white potatoes, most wheat flour products, and most cold cereals, e.g., pumpernickel, rye, multigrain, or sourdough bread, old-fashioned oatmeal, bran cereals, grape-nuts, most fruits, sweet potatoes, pasta, rice, barley, couscous, beans, peas, lentils, most vegetables [Table 1].

Food groups

Choose unlimited amounts of grains, legumes, fruits, and vegetables. Modest amounts of non-fat condiments, alcohol, and coffee are also fine.

Reduce salt (“sodium”) in diet

High blood pressure may also be present with your diabetes. Limiting how much salt you eat can help keep your blood pressure low. Decrease the amount of salt you add during cooking and reduce salt in recipes, before adding salt at the table, taste first, try seasoning your food with (salt-free) herbs, spices, and garlic. Lemon juice brings out the natural saltiness of foods. Avoid processed foods that are high in salt (sodium chloride) such as canned or packaged foods and condiments such as mustard, watch for “Na” (sodium) on food labels. Chips, pretzels, and other such snacks are

very high in salt, and check with your physician before using salt substitutes.

Alcohol

- Limit alcohol to less than 2 drinks per day (1 drink = 12 oz beer = 1.5 oz liquor = 4 oz wine)
- Drinking alcohol is not recommended if you: Have high triglycerides (blood fats), have high blood pressure, have liver problems, are pregnant or breastfeeding.
- If you choose to drink alcohol, remember: To drink with your meal or snack (not on an empty stomach!), to drink slowly or dilute with water or diet soda, that liqueurs, sweet wines and dessert wines have a lot of sugar, to wear your Medic Alert (Alcohol can cause hypoglycemia/low blood glucose), reducing alcohol can promote weight loss and help you lower your blood pressure.

Caffeine

Drink no more than four (4) cups of coffee or caffeine-containing beverages per day.

Food preparation

Avoid adding sugar during cooking. use Splenda (sucralose) instead of sugar if baking or cooking, use low-fat and low sugar sauces and marinades, use reduced-fat cooking methods such as barbecuing, broiling, roasting, and steaming and avoid frying and deep-fat frying.

FOODS TO ALWAYS CHOOSE

Whole grain products

Whole and multi grain breads, whole wheat pasta, brown rice, low-fat and multigrain crackers, low-sugar, whole-grain cereals, oatmeal, bran, bulgur, buckwheat, low-fat, whole grain baked goods with added bran or oat bran [Table 2].

Table 1: Classification of foods on basis of Glycemic index

Glycemic index		
Low-GI (enjoy) or foods <i>choose most often</i>	Medium GI foods <i>choose occasionally</i>	High-GI (avoid) or high GI foods <i>choose less often</i>
Whole grain breads, oatmeal (slow cook oats), all-bran cereal, converted or parboiled rice, brown rice, pumpernickel bread, pasta, all beans, peas and lentils, apple, orange, tomato juice, sweet potato, carrots, broccoli, cauliflower, apples, peaches, pears, grapefruits, oranges, sweet corn, dark leafy vegetables, popcorn, marmalade, jams-small amount, skim or 1% milk, low fat, yogurt, soy beverage	Couscous, basmati rice, shredded wheat cereal, pita bread, rye bread, high fiber crackers baked snack foods (not fried), white bread, whole wheat pretzels, cantaloupe, pineapple	Instant mashed potatoes, baked white potatoes, red potatoes, parsnips, rutabaga, instant rice, corn flakes, rice krispies, refined cereals, bagels, waffles/pancakes, pretzels, soda crackers, rice crackers, French fries, digestive cookies, graham crackers, dried dates, figs, fruit bars, sports drinks, sweetened soft drinks, glucose

Table 2: List of foods to always choose

Vegetables	Fruits	Grains	Legumes
All, except white potatoes. Examples include tomatoes, cucumbers, carrots, broccoli, cauliflower, spinach, kale, collards, squash, green beans, sweet potatoes, and artichokes	All, except avocados, olives, pineapple, and watermelon. bananas, apples, grapes, pears, peaches, oranges, melons, grapefruit, kiwi, and berries, among others, are all good choices	Pasta, rice, high-fiber cereals, corn, oatmeal, couscous, bulgur wheat, millet, barley, rye, etc	Beans (black, pinto, kidney, garbanzo, white, etc.), peas, split peas, lentils, non-fat soy products

Vegetables and fruits

Eat green leafy vegetables; eat an abundance fresh/frozen vegetables and unsweetened, fresh, frozen, or canned fruits

Milk/Dairy products

Dairy products with less than 1% fat and cheese should be 10-20% ME.

Meat and alternatives

Fish (canned in water, fresh, frozen), seafood, skinless chicken and turkey, lean meats with fat trimmed, wild game, lean cold cuts (but watch the salt content), legumes, tofu, eggs (up to 8 per week).

Other foods and fats

Olive, canola, soybean, sesame, sunflower oils (3 tsp or less per day), non-hydrogenated soft margarines, low-calorie dressings and mayonnaise, light peanut butter, nuts (watch salt and calories), unsalted seeds: Flax, pumpkin, sunflower, defatted gravy and low-sugar condiments, cocoa powder or a small piece of dark chocolate.

Sweets (in very small amounts)

Sugar substitutes and artificial sweeteners, low sugar jams/jellies/syrups, sugar-free candies, gelatins, gum, low-sugar and high fiber baked goods, and low-fat and low-sugar frozen dessert

Snack foods

Popcorn without salt, butter, or hydrogenated oils and choose low fat, low sugar snack foods.

ROLE OF PHYSICAL ACTIVITY OR EXERCISE

Regular physical activity helps the body cells take up glucose and thus lower blood glucose levels. Regular physical activity also helps with weight loss as well as controlling blood cholesterol and blood pressure. You need to let your doctor and dietitian know about the kinds of physical activities you do regularly. Your doctor and dietitian will help you balance your physical activity with your medication and diabetic meal plan. If you are not physically active now, your doctor may recommend that you increase physical activity. Important benefits of a regular aerobic exercise program in diabetes management include decreased need for insulin, decreased risk of obesity, and decreased risk for heart disease. Exercise decreases total cholesterol, improves the ratio of low-density lipoprotein (LDL) to high-density lipoprotein cholesterol (HDL), and reduces blood triglycerides. It may also decrease blood pressure and lower stress levels. Walking is one of the easiest and healthiest ways to exercise. This is one activity that anyone can do for a lifetime without special equipment and with little risk of injury. Talk to your doctor about exercise. Supervised activity is best because of the risk of an insulin imbalance. Use the buddy system when you exercise.^[71-77]

Reducing weight

Eat smaller portions of foods and remember that your lunch and dinner plate should be 1/4 protein, 1/4 starch (including

List of foods/drinks to be avoided and their alternatives

Sugary foods/drinks to avoid	Suitable alternatives
Sugar, sucrose, glucose, fructose, dextrose glucose syrup. 'diabetic' sweeteners with sorbitol or other polyols. Sweeteners with some sugars added e.g., hermesetas sprinkle sweet, boots sugarlite, sucron	Artificial sweeteners made with aspartame, saccharin or acesulfame K Tablets, liquid, powder or granulated e.g., canderel, flix, hermesetas, natrena, sweetex
Ordinary fizzy drinks, squashes and cordials e.g., cola, lemonade, lucozade, barley water Sweetened fruit juices	Diet, slimline or one calorie fizzy drinks Sugar free/No added sugar squash. Only a small glass of pure fruit juice with a meal
All sweet biscuits e.g., with syrup, sucrose, chocolate, cream etc.	Plain biscuit e.g., digestive, Hob Nobs, hovis, marie, rich tea, morning coffee, garibaldi, savory crackers, or oatcakes, occasionally
Cakes and sweet pastries	Only occasionally, especially if you are overweight: Whole meal scone, teacake, bara brith, malt loaf or teabread, crumpet or muffin. Home-made low sugar wholemeal cake
Chocolates, sweets, mints, toffees including 'Diabetic' chocolates etc.	Sugar-free mints or chewing gum. Opt for fresh fruit, crispbread or crackers instead
Ordinary and 'Diabetic' jam/marmalade	A little pure fruit, reduced sugar jam, or marmalade
Honey, syrup, treacle, lemon curd	Try savory spreads e.g., a little peanut butter, marmite, or low fat cheese spread on toast or crackers

potatoes), and 1/2 vegetables. Eat 3 balanced meals per day (no more than 6 hours apart), and don't skip meals; snack with fruit between meals. Choose foods lower in fat and sugar; choose low GI index foods whenever possible; avoid "white" foods (white flour and white sugar).

Glucose testing

Your diabetic meal plan, physical activity, and medication are all balanced to help keep your blood glucose levels normal. You need to check your blood glucose levels at home to keep track of how you are doing. Soon you will learn how the foods you eat and your physical activity affect your blood glucose level. The best defense against diabetic complications is to keep blood glucose in control and take good care of yourself. Keeping your blood glucose in control will help you feel better now and stay healthy in the future.^[78-80]

DISCUSSION

The breakfast should be 1/3 fruit, 1/3 starchy fiber foods (multigrain bread and cereal products), and 1/3 protein (nuts, eggs, tofu, beans, lentils, low-fat dairy products). The lunch and dinner plates should be 1/2 vegetables, 1/4 starchy fiber foods, and 1/4 protein. Choose whole grains, such as whole wheat

pasta, whole wheat bread, and brown rice to increase fiber intake. Most of these are low in fat. Choose only lean meat and poultry.^[81-84] Remove skin and trim fat before cooking (50-100 g or 2-4 oz). See the milk fat (MF) of all dairy products. Use skim or 1% milk products and low-fat cheese (less than 20% MF), or choose fortified soy products. Reduce your total fat intake (less than 25% - 35% of your daily calories). To achieve this, always try to choose low fat foods and avoid fried foods. Limit saturated and trans fats to less than 10% of your daily calories. Try to always choose unsaturated fats such as olive and canola oils and non-hydrogenated margarine (in moderation). Saturated and trans fats raise blood cholesterol levels, while unsaturated fats lower blood cholesterol. Saturated fats are solid at room temperature and are usually of animal origin. They are found in meats, whole milk, dairy products, butter, and hard margarines.^[85-90] Trans fats are found in baked and pre-packaged foods. Hydrogenation is a process that changes liquid vegetable oil into a solid fat such as hard margarine. The hydrogenation process changes some of the good fats into cholesterol-raising saturated and trans fats. People with diabetes are at a greater risk of developing or have already high levels of fats in their heart and blood vessels. Omega-3 fatty acids are found in cold water fish such as herring, mackerel, salmon, trout, sardines and tuna, and in flaxseeds (2 tbsp per day, freshly ground).^[90-93] Three to four servings of fish per week is recommended as part of a healthy, balanced diet. Omega-enriched foods are also available in supermarkets such as omega-3 eggs and omega-3 enriched dairy products. Omega-3 supplements: Always look for the active ingredients DHA and EPA. Recommendations are 600-900 mg/day. Always check with your doctor or registered dietitian before taking any supplements. Increase fiber in your diet by eating more whole grain foods, vegetables, fruits, and legumes.^[94-96] These foods also contain vitamins, minerals, and antioxidants and have a lower glycemic index. Low glycemic index foods will help to keep your blood sugar levels in the target range.^[97-99]

CONCLUSION

In conclusion, effective lifestyle modifications including counseling on weight loss, adoption of a healthy dietary pattern like the Mediterranean diet, together with physical activity are the cornerstone in the prevention of type-2 diabetes. Therefore, emphasis must be given to promoting a healthier lifestyle and finding solutions in order to increase adherence and compliance to the lifestyle modifications, especially for high-risk individuals. Results from epidemiological studies and clinical trials evaluating the role of the Mediterranean dietary pattern regarding the development and treatment of type-2 diabetes indicate the protective role of this pattern. As a result, promoting adherence to the Mediterranean diet is of considerable public health importance as this dietary pattern, apart from its various health benefits, is tasty and easy to follow in the long-term. Diet is an important aspect in the management of a diabetic patient. The diabetic healthcare provider and the patient should understand the basic dietary needs of the patient. In this form, there may be plenty of insulin in the bloodstream, but the cells are resistant to it. Glucose cannot easily get into the cells, and it backs up in

the bloodstream. Over the short run, people with uncontrolled diabetes may experience fatigue, thirst, frequent urination, and blurred vision. In the long run, they are at risk for heart disease, kidney problems, disorders of vision, nerve damage, and other difficulties.

There is no cure for diabetes. However, you can manage or delay diabetes through diet, exercise, weight control and, if necessary, medication.

-Jacquelyn W. McClelland

REFERENCES

1. Wild S, Roglic G, Green A, Sicree R, King H. Global prevalence of diabetes: Estimates for the year 2000 and projections for 2030. *Diabetes Care* 2004;27:1047-53.
2. Giugliano D, Esposito K. Mediterranean diet and metabolic diseases. *Curr Opin Lipidol* 2008;19:63-8.
3. Martinez-Gonzalez MA, de la Fuente-Arillaga C, Nunez-Cordoba JM, Basterra-Gortari FJ, Beunza JJ, Vazquez Z, et al. Adherence to Mediterranean diet and risk of developing diabetes: Prospective cohort study. *BMJ* 2008;336:1348-51.
4. Sanchez-Tainta A, Estruch R, Bullo M, Corella D, Gomez-Gracia E, Fiol M. Adherence to a Mediterranean-type diet and reduced prevalence of clustered cardiovascular risk factors in a cohort of 3,204 high-risk patients. *Eur J Cardiovasc Prev Rehabil* 2008;15:589-93.
5. Panagiotakos DB, Pitsavos C, Chrysoschoou C, Stefanadis C. The epidemiology of Type 2 diabetes mellitus in Greek adults: The ATTICA study. *Diabet Med* 2005;22:1581-8.
6. World Health Organization. Definition, Diagnosis and Classification of Diabetes Mellitus and Its Complications. Part 1. Report of WHO consultation. Geneva: WHO; 1999.
7. Prevention and Management of the Global Epidemic of Obesity. Report of the WHO consultation on obesity, Geneva: WHO; 1998.
8. Otuyelu F. Diabetic diet for the Nigerian. *Niger Med Pract* 1982;3:48-51.
9. Fadupin GT, Keshinro OO. Factors influencing dietary compliance and glycaemic control in adult diabetic patients in Nigeria. *Diabetes Int* 2001;11:59-61.
10. Fadupin GT, Keshinro OO, Sule ON. Dietary recommendations: Example of advice given to diabetic patients in Nigeria. *Diabetes Int* 2000;10:68-70.
11. Sofi F, Innocenti G, Dini C, Masi L, Battistini NC, Brandi ML, et al. Low adherence of a clinically healthy Italian population to nutritional recommendations for primary prevention of chronic diseases. *Nutr Metab Cardiovasc Dis* 2006;16:436-44.
12. Kastorini CM, Panagiotakos DB. Mediterranean diet and diabetes prevention: Myth or fact? *World J Diabetes* 2010;1:65-7.
13. Colditz GA, Willett WC, Rotnitzky A, Manson JE. Weight gain as a risk factor for clinical diabetes mellitus in women. *Ann Intern Med* 1995;122:481-6.
14. Feskens EJ, van Dam RM. Dietary fat and the etiology of type 2 diabetes: An epidemiological perspective. *Nutr Metab Cardiovasc Dis* 1999;9:87-95.
15. Bergman RN, Ader L. Free fatty acids and the pathogenesis of type 2 diabetes mellitus. *Trends Endocrinol Metab* 2001;3(suppl):S11-9.
16. Panagiotakos DB, Polystipioti A, Papairakleous N, Polychronopoulos E. Long-term adoption of a Mediterranean diet is associated with a better health status in elderly people; a cross-sectional survey in Cyprus. *Asia Pac J Clin Nutr* 2007;16:331-7.
17. Panagiotakos DB, Tzima N, Pitsavos C, Chrysoschoou C, Zampelas A, Tousoulis D, et al. The association between adherence to the Mediterranean diet and fasting indices of glucose homeostasis: The Attica Study. *J Am Coll Nutr* 2007;26:32-8.
18. American Diabetes Association. Standards of medical care in

- diabetes-2010. *Diabetes Care* 2010;33(suppl 1):S11-61.
19. American Diabetes Association: Evidence-based nutrition principles and recommendations for the treatment and prevention of diabetes and related complications (Position Statement). *Diabetes Care* 2003;26(Suppl 1):S51-61.
20. Barnard ND, Scialli AR, Turner-McGrievy G, Lanou AJ, Glass J. The effects of a lowfat, plant-based dietary intervention on body weight, metabolism, and insulin sensitivity. *Am J Med* 2005;118:991-7.
21. American Diabetes Association. Standards of Medical Care in Diabetes—2008. *Diabetes Care* 2008;31:S12-S54.
22. Barnard ND, Scialli AR, Turner-McGrievy GM, Lanou AJ. Acceptability of a low-fat vegan diet compares favorably to a step II diet in a randomized, controlled trial. *J Cardiopulm Rehabil* 2004;24:229-35.
23. Barnard RJ, Jung T, Inkeles SB. Diet and exercise in the treatment of NIDDM: The need for early emphasis. *Diabetes Care* 1994;17:1-4.
24. Wieland H, Seidel D. A simple specific method for precipitation of low density lipoproteins. *J Lipid Res* 1983;24:904-9.
25. Yang W, Lu J, Weng J, Jia W, Ji L, Xiao J, *et al.* Prevalence of diabetes among men and women in China. *N Engl J Med* 2010;362:1090-101.
26. Holt SH, Brand Miller JC, Petocz P, Farmakaladis E. A satiety index of common foods. *Eur J Clin Nutr* 1995;49:675-90.
27. Nicholson AS, Sklar M, Barnard ND, Gore S, Sullivan R, Browning S. Toward improved management of NIDDM: A randomized, controlled, pilot intervention using a low-fat, vegetarian diet. *Prev Med* 1999;29:87-91.
28. UK Prospective Diabetes Group. Intensive bloodglucose control with sulphonylureas and insulin compared with conventional treatment and risk of complications in patients with type 2 diabetes (UKPDS 33). *Lancet* 1998;352:837-53.
29. Wolf AM, Conaway MR, Crowther JQ, Hazen KY, L Nadler J, Oneida B, *et al.* Translating lifestyle intervention to practice in obese patients with type 2 diabetes: Improving control with activity and nutrition (ICAN) study. *Diabetes Care* 2004;27:1570-6.
30. The Diabetes Control and Complications Trial Research Group. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. *N Engl J Med* 1993;329:977-86.
31. The International Expert Committee. International Expert Committee report on the role of the A1C assay in the diagnosis of diabetes. *Diabetes Care* 2009;32:1327-34.
32. Sinitskaya N, Gourmelen S, Schuster-Klein C, Guardiola-Lemaitre B, Pevet P, Challet E. Increasing fat-to-carbohydrate ratio in a high-fat diet prevents the development of obesity but not a prediabetic state in rats. *Clin Sci* 2007;113:417-25.
33. Sjostrom CD, Peltonen M, Wedel H, Sjostrom L. Differentiated long-term effects of intentional weight loss on diabetes and hypertension. *Hypertension* 2000;36:20-5.
34. Schroder H. Protective mechanisms of the Mediterranean diet in obesity and type 2 diabetes. *J Nutr Biochem* 2007;18:149-60.
35. Selvin E, Crainiceanu CM, Brancati FL, Coresh J. Short-term variability in measures of glycemia and implications for the classification of diabetes. *Arch Intern Med* 2007;167:1545-51.
36. Shai I, Schwarzfuchs D, Henkin Y, Shahar DR, Witkow S, Greenberg I, *et al.* Weight loss with a lowcarbohydrate, Mediterranean, or low-fat diet. *N Engl J Med* 2008;359:229-41.
37. Lindeberg S, Jönsson T, Granfeldt Y, Borgstrand E, Soffman J, Sjöström K, *et al.* A Palaeolithic diet improves glucose tolerance more than a Mediterranean-like diet in individuals with ischaemic heart disease. *Diabetologia* 2007;50:1795-807.
38. Lovejoy JC, Windhauser MM, Rood JC, de la Bretonne JA. Effect of a controlled high-fat versus low-fat diet on insulin sensitivity and leptin levels in African-American and Caucasian women. *Metabolism* 1998;47:1520-4.
39. American Diabetes Association. Clinical practice recommendations-2007. *Diabetes Management in Correctional Institutions: Diabetes Care* 2007;30(Suppl 1):S77-84.
40. American Diabetes Association. Diagnosis and classification of diabetes mellitus. *Diabetes Care* 2010;33(Suppl 1):S62-9.
41. Bouchard C, Tremblay A, LeBlanc C, Lortie G, Savard R, Theriault G. A method to assess energy expenditure in children and adults. *Am J Clin Nutr* 1983;37:461-7.
42. Drewnowski A. Energy density, palatability, and satiety: Implications for weight control. *Nutr Rev* 1998;56:347-53.
43. Duncan KH, Bacon JA, Weinsier RL. The effects of high and low energy density diets on satiety, energy intake, and eating time of obese and nonobese subjects. *Am J Clin Nutr* 1983;37:763-7.
44. Eaton SB, Eaton SB, Konner MJ, Shostak M. An evolutionary perspective enhances understanding of human nutritional requirements. *J Nutr* 1996;126:1732-40.
45. Esposito K, Maiorino MI, Di Palo C, Giugliano D. Adherence to a Mediterranean diet and glycaemic control in Type 2 diabetes mellitus. *Diabet Med* 2009;26:900-7.
46. Franz MJ, Splett PL, Monk A, Barry B, McClain K, Weaver T, *et al.* Cost-effectiveness of medical nutrition therapy provided by dietitians for persons with non-insulin-dependent diabetes mellitus. *J Am Diet Assoc* 1995;95:1018-24.
47. Friedewald WT, Levy RI, Fredrickson DS. Estimation of the concentration of lowdensity lipoprotein cholesterol in plasma without use of the preparative ultracentrifuge. *Clin Chem* 1972;18:499-502.
48. Garg A, Bantle JP, Henry RR, Coulston AM, Griver KA, Raatz SK, *et al.* Effects of varying carbohydrate content of the diet in patients with non-insulin-dependent diabetes mellitus. *JAMA* 1994;271:1421-8.
49. Garg A. High-monosaturated fat diets for patients with diabetes mellitus: A meta-analysis. *Am J Clin Nutr* 1998;67:577S-82.
50. Howarth NC, Saltzman E, Roberts SB. Dietary fiber and weight regulation (Review). *Nutr Rev* 2001;59:129-39.
51. Pérez-Jiménez F, López-Miranda J, Pinillos MD, Gómez P, Paz-Rojas E, Montilla P, *et al.* A Mediterranean and a high-carbohydrate diet improve glucose metabolism in healthy young persons. *Diabetologia* 2001;44:2038-43.
52. Schakel SF, Sievert YA, Buzzard IM. Sources of data for developing and maintaining a nutrient database. *J Am Diet Assoc* 1988;88:1268-71.
53. Simopoulos AP. The Mediterranean diets: What is so special about the diet of Greece? The scientific evidence. *J Nutr* 2001;131:3065S-73.
54. Wilson DH, Bogacz JP, Forsythe CM, Turk PJ, Lane TL, Gates RC, *et al.* Fully automated assay of glycohemoglobin with the Abbott IMx analyzer: Novel approaches for separation and detection. *Clin Chem* 1993;39:2090-7.
55. Weiss R, Dufour S, Taksali SE, Tamborlane WV, Petersen KF, Bonadonna RC, *et al.* Prediabetes in obese youth: A syndrome of impaired glucose tolerance, severe insulin resistance, and altered myocellular and abdominal fat partitioning. *Lancet* 2003;362:951-7.
56. Sparks LM, Xie H, Koza RA, Mynatt R, Hulver MW, Bray GA, *et al.* A highfat diet coordinately downregulates genes required for mitochondrial oxidative phosphorylation in skeletal muscle. *Diabetes* 2005;54:1926-33.
57. Centers for Disease Control and Prevention. Transmission of hepatitis B virus among persons undergoing blood glucose monitoring in long-term-care facilities—Mississippi, North Carolina, and Los Angeles County, California, 2003–2004. *MMWR Morb Mortal Wkly Rep* 2005;54:220-3.
58. Estruch R, Martínez-González MA, Corella D, Salas-Salvadó J, Ruiz-Gutiérrez V, Covas MI, *et al.* Effects of a Mediterranean-style diet on cardiovascular risk factors: A randomized trial. *Ann Intern Med* 2006;145:1-11.
59. Federal Bureau of Prisons. Clinical practice guidelines: Preventive health care. Federal Bureau of Prisons, Washington, DC: Federal Bureau of Prisons; 2007.
60. Goff LM, Bell JD, So PW, Dornhorst A, Frost GS. Veganism and its relationship with insulin resistance and intramyocellular lipid. *Eur J Clin Nutr* 2005;59:291-8.
61. Centers for Disease Control and Prevention. National Diabetes Fact Sheet: General Information and National Estimates on Diabetes

- in the United States, 2007. Atlanta, GA: US Department of Health and Human Services, Centers for Disease Control and Prevention; 2008. [Last accessed on 2010 May 18].
62. Field AE, Willett WC, Lissner L, Colditz GA. Dietary fat and weight gain among women in the Nurses' Health Study. *Obesity* 2007;15:967-6.
 63. Ford ES, Giles WH, Dietz WH. Prevalence of the metabolic syndrome among US adults. Findings from the Third National Health and Nutrition Survey. *JAMA* 2001;287:356-9.
 64. Fraser GE. Vegetarianism and obesity, hypertension, diabetes, and arthritis. In *Diet, Life Expectancy, and Chronic Disease*. Oxford UK; Oxford University Press; 2003. p. 129-48.
 65. Trichopoulos A. Mediterranean diet: The past and the present. *Nutr Metab Cardiovasc Dis* 2001;11:1-4.
 66. US Preventive Services Task Force. Screening for type 2 diabetes mellitus in adults: US Preventive Services Task Force Recommendation Statement. *Ann Intern Med* 2008;148:846-54.
 67. Nathan DM, Kuenen J, Borg R, Zheng H, Schoenfeld D, Heine RJ. for the A1C-derived Average Glucose (ADAG) Study Group. Translating the A1C assay into estimated average glucose values. *Diabetes Care* 2008;31:1473-8.
 68. Pan XR, Li GW, Hu YH, Wang JX, Yang WY, An ZX, *et al.* Effects of diet and exercise in preventing NIDDM in people with impaired glucose tolerance: The Da Qing IGT and Diabetes Study. *Diabetes Care* 1997;20:537-44.
 69. Patel RB, Burke TF. Urbanization: An emerging humanitarian disaster. *N Engl J Med* 2009;361:741-3.
 70. Petersen KF, Dufour S, Befroy D, Garcia R, Shulman GI. Impaired mitochondrial activity in the insulin-resistant offspring of patients with type 2 diabetes. *N Engl J Med* 2004;350:664-71.
 71. Qi L, Hu FB, Hu G. Genes, environment, and interactions in prevention of type 2 diabetes: A focus on physical activity and lifestyle changes. *Curr Mol Med* 2008;8:519-32.
 72. Abelson R. An insurer's new approach to diabetes. *New York: The New York Times*; 2010. p. 14.
 73. Hu FB. Dietary fat and meat intake in relation to risk of type 2 diabetes in men. *Diabetes Care* 2002;25:417-24.
 74. Hua NW, Stoohs RA, Facchini FS. Low iron status and enhanced insulin sensitivity in lacto-ovo vegetarians. *Br J Nutr* 2001;86:515-9.
 75. Kendall A, Levitsky DA, Strupp BJ, Lissner L. Weight loss on a low-fat diet: Consequence of the imprecision of the control of food intake in humans. *Am J Clin Nutr* 1991;53:1124-9.
 76. Kleges RC, Kleges LM, Haddock CK, Eck LH. A longitudinal analysis of the impact of dietary intake and physical activity on weight change in adults. *Am J Clin Nutr* 1992;55:818-22.
 77. Rolls BJ. The role of energy density in the over consumption of fat. *J Nutr* 2000;130:268S-71.
 78. Zammit VA, Waterman IJ, Topping D, McKay G. Insulin stimulation of hepatic triacylglycerol secretion and the etiology of insulin resistance. *J Nutr* 2001;131:2074-7.
 79. Hossain P, Kavar B, El Nahas M. Obesity and diabetes in the developing world-a growing challenge. *N Engl J Med* 2007;356:213-5.
 80. Ackerman RT, Finch EA, Brizendine E, Zhou H, Marrero DG. Translating the Diabetes Prevention Program into the community. The Deploy Pilot Study. *Am J Prev Med* 2008;35:357-63.
 81. Allain CC, Poon LS, Chan CS, Richmond W, Fu PC. Enzymatic determination of total serum cholesterol. *Clin Chem* 1974;20:470-5.
 82. Gaede PH, Lund-Andersen H, Parving HH, Pedersen O. Effect of a multifactorial intervention on mortality in type 2 diabetes. *N Engl J Med* 2008;358:580-91.
 83. Guldstrand M, Adamson U, Ahlren B. Improved β -cell function after standardized weight reduction in severely obese subjects. *Am J Physiol Endocrinol Metab* 2003;284:E557-65.
 84. Haber GB, Heaton KW, Murphy D, Burroughs LF. Depletion and disruption of dietary fibre: Effects on satiety, plasma-glucose, and serum-insulin. *Lancet* 1977;2:679-82.
 85. Laitinen JH, Ahola IE, Sarkkinen ES, Winberg RL, Harmaakorpi-livonen PA, Uusitupa MI. Impact of intensified dietary therapy on energy and nutrient intakes and fatty acid composition of serum lipids in patients with recently diagnosed noninsulin- dependent diabetes mellitus. *J Am Diet Assoc* 1993;93:276-83.
 86. Mattson JS, Cerutis RD. Diabetes mellitus: A review of the literature and dental implications. *Compend Contin Educ Dent* 2001;22:757-70.
 87. Mokdad AH. The continuing epidemics of obesity and diabetes in the United States. *JAMA* 2001;286:1195-200.
 88. Morgan SA, O'Dea K, Sinclair AJ. A low-fat diet supplemented with monounsaturated fat results in less HDL-C lowering than a very-low-fat diet. *J Am Diet Assoc* 1997;97:151-6.
 89. Jenkins DJ, Kendall CW, Marchie A, Faulkner DA, Wong JM, de Souza R, *et al.* Effects of a dietary portfolio on cholesterol-lowering foods vs lovastatin on serum lipids and C-reactive protein. *JAMA* 2003;290:502-10.
 90. Jenkins DJ, Kendall CW, Marchie A, Jenkins AL, Augustin LS, Ludwig DS, *et al.* Type 2 diabetes and the vegetarian diet. *Am J Clin Nutr* 2003;78:610S-6.
 91. Krentz AJ, Bailey CJ. Oral antidiabetic agents: Current role in type 2 diabetes mellitus (Review). *Drugs* 2005;65:385-411.
 92. Asif M. The role of fruits, vegetables, and spices in diabetes. *Int J Nutr Pharmacol Neurol Dis* 2011;1:26-34.
 93. Roberts CK, Vaziri ND, Barnard RJ. Effect of diet and exercise intervention on blood pressure, insulin, oxidative stress, and nitric oxide availability. *Circulation* 2002;106:2530-2.
 94. Stubbs RJ, Johnstone AM, Harbron CG, Reid C. Covert manipulation of energy density of high carbohydrate diet in 'pseudo free-living' humans. *Int J Obes* 1998;22:885-92.
 95. Tremblay A. Nutritional determinants of the insulin resistance syndrome. *Int J Obes* 1995;19(suppl):S60-5.
 96. Vrieze A, Holleman F, Zoetendal EG, de Vos WM, Hoekstra JB, Nieuwdorp M. The environment within: How gut microbiota may influence metabolism and body composition. *Diabetologia* 2010;53:606-13.
 97. Wieland H, Seidel D. A simple specific method for precipitation of low density lipoproteins. *J Lipid Res* 1983;24:904-9.
 98. Yang W, Lu J, Weng J, Jia W, Ji L, Xiao J, *et al.* Prevalence of diabetes among men and women in China. *N Engl J Med* 2010;362:1090-101.
 99. Holt SH, Brand Miller JC, Petocz P, Farmakaladis E. A satiety index of common foods. *Eur J Clin Nutr* 1995;49:675-90.

Source of Support: Nil, **Conflict of Interest:** None declared